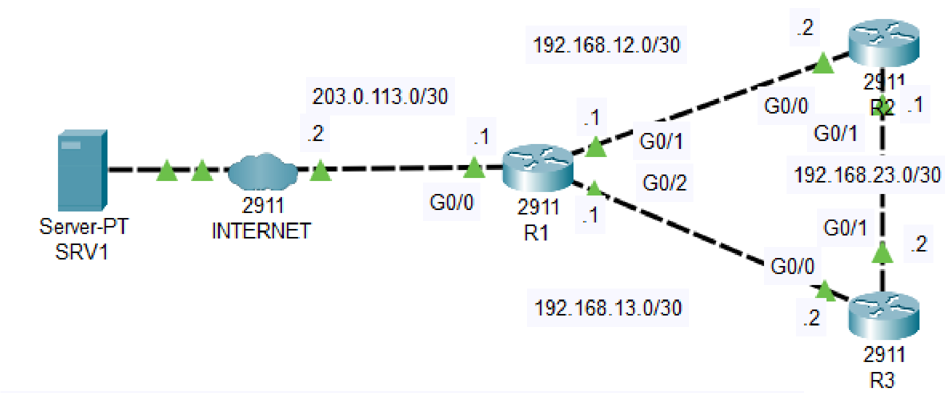

Topology

ROUTING HAS BEEN PRECONFIGURED

(default route on R1, OSPF on all routers with 'network 0.0.0.0 255.255.255.255 area 0')

1. **Configure the software clock on R1, R2, and R3 to 12:00:00 Dec 30 2020 (UTC).**

```
R1>en
R1#clock set 12:00:00 Dec 30 2020
R1#show clock detail
12:0:4.364 UTC Wed Dec 30 2020
Time source is user configuration
R2>en
R2#clock set 12:00:00 Dec 30 2020
R3>en
R3#clock set 12:00:00 Dec 30 2020
```

   NTP can take quite a while to synchronize. But if the time on the device is manually configured to the time of the NTP server, the synchronization process will be quicker.

2. **Configure the time zone of R1, R2, and R3 to match your own.**

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#clock timezone PST 8
R1(config)#do show clock detail
20:3:5.335 PST Wed Dec 30 2020
Time source is user configuration
```

   Philippine time is offset by +8 hours from UTP.

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#clock timezone PST 8
R2(config)#do show clock detail
20:0:13.43 PST Wed Dec 30 2020
Time source is user configuration
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#clock timezone PST 8
R3(config)#do show clock detail
20:0:8.713 PST Wed Dec 30 2020
Time source is user configuration
```

   The clocks on R1, R2, and R3 don't match at this moment. This will be fixed upon enabling NTP.

3. **Configure R1 to synchronize to NTP server 1.1.1.1 over the Internet. What stratum is 1.1.1.1? What stratum is R1?**

```
R1(config)#ntp server 1.1.1.1
R1(config)#do sh ntp associations

address         ref clock       st   when     poll    reach  delay          offset            disp
 ~1.1.1.1       .INIT.          16   -        64      0      0.00           0.00              0.01
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R1(config)#do sh ntp associations

address         ref clock       st   when     poll    reach  delay          offset            disp
*~1.1.1.1       .INIT.          0    10       32      377    0.00           0.00              0.12
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R1#show ntp status
Clock is synchronized, stratum 1, reference is 1.1.1.1
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is E36A0DED.0000007F (11:20:13.127 UTC Wed Dec 30 2020)
clock offset is 0.00 msec, root delay is 0.00  msec
root dispersion is 18.09 msec, peer dispersion is 0.24 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll
interval is 6, last update was 26 sec ago.
R1#show clock detail
19:20:55.551 PST Wed Dec 30 2020
Time source is NTP
```

The sys.peer asterisk initially does not show up on the configured NTP server. This can take quite a while.

4. **Configure R1 as a stratum 8 NTP master. Synchronize R2 and R3 to R1 with authentication.*the 'ntp source' command is not available in Packet Tracer, so just use the physical interface IP addresses of R1.**

```
R1(config)#ntp master
```

Since 8 is the default NTP stratum, no need to specify the stratum number.

```
R1(config)#ntp authenticate
R1(config)#ntp authentication-key 1 md5 mylab
R1(config)#ntp trusted-key 1
R2(config)#ntp authenticate
R2(config)#ntp authentication-key 1 md5 mylab
R2(config)#ntp trusted-key 1
R2(config)#ntp server 192.168.12.1 key 1
```

The NTP source command isn't available in packet tracer, so instead of using a loopback interface, just configured R1's G0/1 interface as the NTP server. The reason you need the NTP source command is if you configure the R1's loopback interface as R2's NTP server, but the replies from R1 come from the IP address on the physical interface, R2 won't sync to R1.

```
R2#show ntp associations

address         ref clock       st   when     poll    reach  delay          offset            disp
*~192.168.12.1  1.1.1.1         1    12       64      377    0.00           0.00              0.48
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

After some time, it has synced to R1.

```
R2#show clock detail
19:43:12.315 PST Wed Dec 30 2020
Time source is NTP
R3(config)#ntp authenticate
R3(config)#ntp authentication-key 1 md5 mylab
R3(config)#ntp trusted-key 1
R3(config)#ntp server 192.168.13.1 key 1
```

```
R3(config)#do sh ntp associations

address         ref clock       st   when     poll    reach  delay          offset            disp
*~192.168.13.1  1.1.1.1         1    3        32      377    0.00           0.00              0.24
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

After some time, it has synced to R1.

5. **Configure NTP to update the hardware calendars of R1, R2, and R3.*you can't view the calendar in Packet Tracer**

```
R2(config)#ntp update-calendar
```

```
R3(config)#ntp update-calendar
```